Robustness across conditions

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Keep condition-specific results visible when checking robustness

Compare frozen procedures across declared conditions without replacing slices with one pooled score.

Keep one habit: Question and contract → complete main ledger → separate picture → boundary probes → investigation → independent attempt.

MAIN LESSON · SECTION 1

Keep condition-specific results visible when checking robustness

> Compare frozen procedures across declared conditions without replacing slices with one pooled score.

Learning goals

I can explain the **mechanism** and terminology of robustness across conditions.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Retrieve before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Inspecting failures and anomalies. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Reproducible environment and records. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Predefining an analysis plan. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Starting explanation

- Recall Inspecting failures and anomalies. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

End goal

Explain robustness across conditions using the records and boundaries of this fictional investigation.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

This week you will practise:

Explain the **mechanism** and terminology of robustness across conditions

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

This week's end goal

MAIN LESSON · SECTION 2

Understand the question and evidence contract

> Compare frozen procedures across declared conditions without replacing slices with one pooled score.

Start with the research problem

A method can do well in one setting and poorly in another. Robustness checks ask how the frozen procedure behaves across relevant declared conditions, using aligned scoring, eligibility and baselines. Report each slice and its denominator before any overall total. A difference identifies a pattern to investigate, not its cause.

A pooled score can conceal a weak condition, especially when the mix of cases differs from the intended use.

Original concept explanation

Robustness checks examine a frozen procedure under declared varied conditions. Performance differences do not by themselves identify cause.

Complete case inputs, rules and limits

Robustness item	Complete supplied exercise contract
Main condition counts	Current sources: 9/10 pass; outdated sources: 5/10
Main interpretation	Condition-associated descriptive difference; no causal mechanism identified
Separate picture	Frozen classifier and rubric unchanged; indoor 7/8 correct, outdoor 3/8
Picture pooled count	Ten correct among sixteen cases
Needed for a controlled follow-up	Aligned baseline, scoring and eligibility; recorded case/condition changes
Not supplied	Per-case outputs, matched-case transition identities, stable population rates or significance
Scope	Fictional supplied summaries; no new model run or data collection

Name given and missing evidence

List the given facts and one fact that is not supplied.

▼ **Earlier wording, alternatives and picture transcripts**

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

MAIN LESSON · SECTION 3

Trace the complete main case before interpreting it

> Follow each evidence-to-conclusion step and retain unknown quantities.

Original declared procedure

Keep baseline, metric and eligibility aligned across conditions. Report condition counts and failures separately; new conditions may extend **evidence** but cannot guarantee all conditions.

Complete main-case evidence ledger

Main quantity	Current sources	Outdated sources
Passes / evaluated	9/10	5/10
Observed rate	90%	50%
Current minus outdated	40 percentage points	Lower quality associated with this condition here
Mechanism	Not identified	Inspect source validity and procedure traces
Generalization	Only supplied cases	No universal causal or robustness conclusion

State the scope and explain the visual

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

Robustness across conditions — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Method passes 9/10 on current sources and 5/10 on outdated sources.	Keep baseline, metric and eligibility aligned across conditions. Report condition counts and failures separately; new conditions may extend evidence but cannot guarantee all conditions.	90% versus 50%; gap 40 percentage points. The outdated-source condition is associated with lower quality here; inspect mechanisms before general causal claims.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Predict the changed case before reading the explanation

> Keep the rule and evidence boundary visible.

Main task and response

- Method passes 9/10 on current sources and 5/10 on outdated sources. Show the intermediate steps and give the conclusion with its scope.

Keep the procedure fixed. Explain what changed in the inputs and result.

Predict the pictured change

- Change: Report only the combined 10/16=62.5%.

▼ **Check after predicting**

Prediction: The arithmetic is correct, but the condition gap becomes hidden.

Reason: Combining records discards the slice distinction relevant to the deployment goal.

▼ **Earlier wording, alternatives and picture transcripts**

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Connect the main result with the separate pictured case

> Keep the two scenarios and their supplied evidence distinct.

Main case and original checked reasoning

Method passes 9/10 on current sources and 5/10 on outdated sources.

90% versus 50%; gap 40 percentage points. The outdated-source condition is associated with lower quality here; inspect **mechanism**s before general causal claims.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Separate pictured inputs and complete explanation

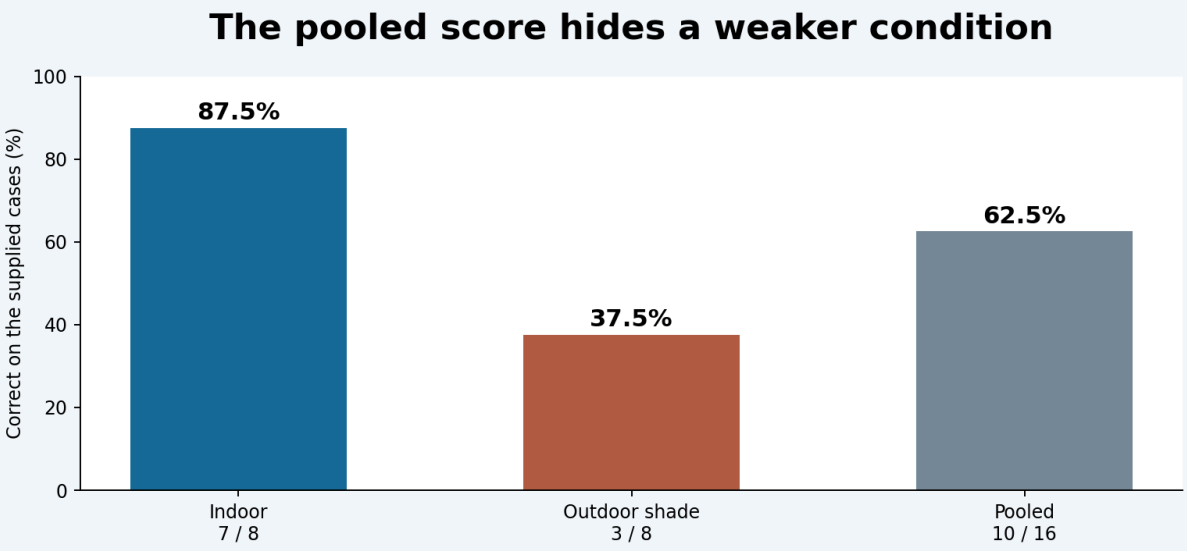
Check robustness across conditions. A fixed classifier is tested separately on indoor and shaded outdoor cases.

Indoor 7/8 correct; outdoor 3/8 correct. The rubric and frozen model are unchanged.

The separate frozen-classifier picture has seven correct of eight indoor cases, or 87.5%, and three correct of eight shaded outdoor cases, or 37.5%. The descriptive slice gap is fifty percentage points. Combined correctness is ten of sixteen, or 62.5%. Reporting only that pooled score hides the weaker outdoor slice and is not a result for either setting alone.

Eight cases per slice do not establish stable population rates or the reason for the difference. Keep the model and rubric frozen in the stated comparison, but do not mistake that exercise assumption for code-verified controls or full sampling evidence.

Read the labelled representation



Same frozen model and rubric stipulated. Slice gap: 50 percentage points.
Eight cases per slice do not establish cause, stable population rates or all-condition robustness.

AI ACADEMY / AI-7/27 · 02 · TRACE THE CASE

Follow the deciding steps



01 Name both slices

Each condition has eight labelled cases.



02 Compute rates

Indoor is 87.5%; outdoor is 37.5%.



03 Bound robustness

The large observed gap motivates investigation rather than claiming condition independence.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Name both slices	Each condition has eight labelled cases.
2. Compute rates	Indoor is 87.5%; outdoor is 37.5%.
3. Bound robustness	The large observed gap motivates investigation rather than claiming condition independence.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Robustness asks whether behaviour survives relevant changes in conditions. A total score can hide a weak slice; keep the scenario and denominator visible.

Complete picture evidence and limits

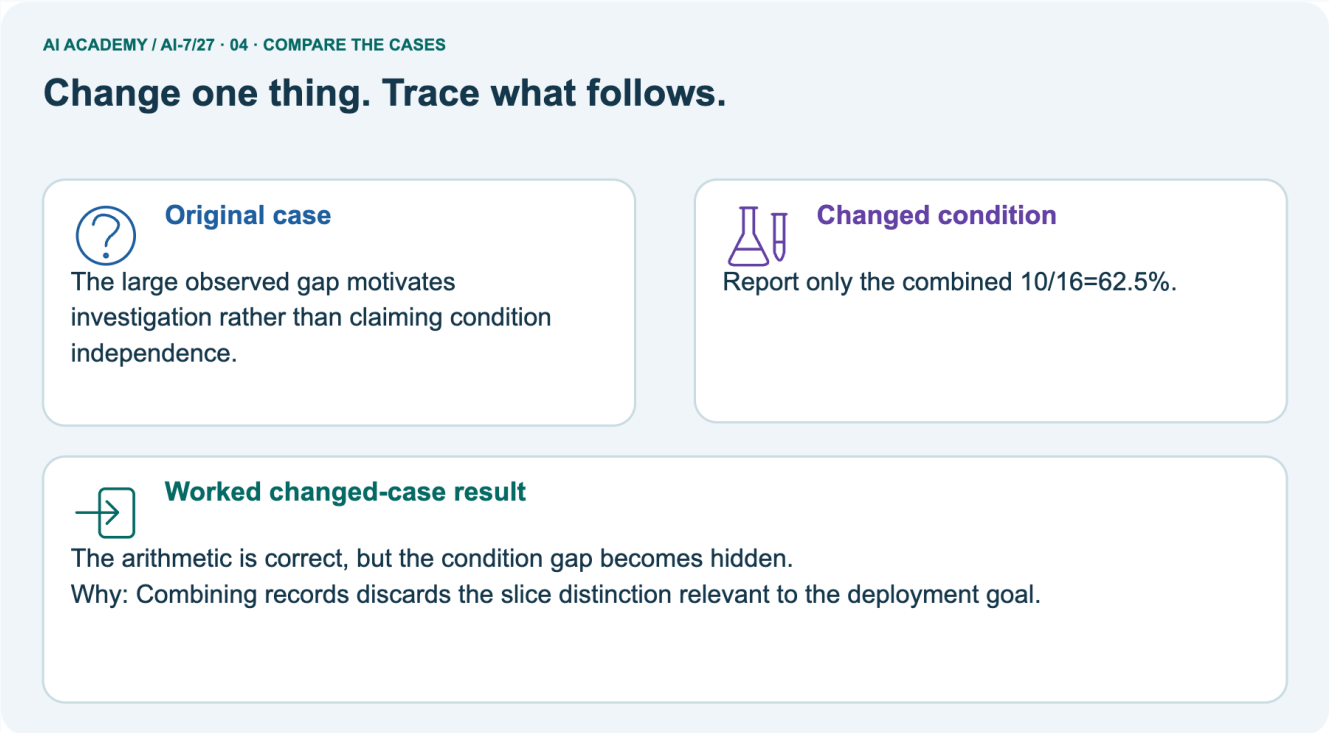
Separate picture slice	Correct / total	Observed rate
Indoor	7/8	87.5%
Outdoor shade	3/8	37.5%
Pooled	10/16	62.5%
Indoor minus outdoor	Different supplied slices	50 percentage points
Hidden by pooled-only report	Outdoor weakness	No separate outdoor rate visible

Separate picture slice	Correct / total	Observed rate
Why rates differ	Not supplied	No causal or significance claim

Predict the changed condition

- Change one thing: Report only the combined 10/16=62.5%. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain what remains unknown

- Eight cases per slice do not establish stable population rates or the reason for the difference.

- Read the labelled picture: Calculate indoor, outdoor and combined rates. Which relevant weakness disappears from view when only the combined rate is reported?

▼ **Earlier wording, alternatives and picture transcripts**

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Robustness across conditions

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/27

Robustness across conditions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Name both slices

Each condition has eight labelled cases.



2. Compute rates

Indoor is 87.5%; outdoor is 37.5%.



3. Bound robustness

The large observed gap motivates investigation rather than claiming condition independence.

CHANGE ONE THING

Report only the combined 10/16=62.5%.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Find the first unsupported inference and repair it

> Use the supplied conditions rather than inventing missing evidence.

Original qualification and repair setup

Robustness is bounded by the tested variations and independent **evidence** policy.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Complete boundary and misconception probes

Robustness audit	Checked interpretation
Overall score looks acceptable	A weak relevant condition can still be hidden
Same frozen model and rubric	Useful declared controls, not proof all case properties align
Outdoor score lower	Descriptive pattern, not a root-cause diagnosis
New condition tested	Extends evidence to that condition only
One candidate improves one slice	Check whether another slice worsens
Two conditions pooled	Their observed weights may differ from intended use
Only aggregate counts supplied	Cannot reconstruct individual paired flips
No statistical analysis supplied	Do not label slice gaps statistically established

Write and justify the repair

- A report omits this qualification: Robustness is bounded by the tested variations and independent **evidence** policy. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise the mechanism with progressively less help

Keep the selected independent case for a first attempt.

Shared explanation

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Supported workbook trace

Use workbook W2 for the connected example “Robustness across conditions”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual scope check

- Use the worked case for Robustness across conditions. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Picture response

- Calculate indoor, outdoor and combined rates. Which relevant weakness disappears from view when only the combined rate is reported?

> **My explanation using a pictured step and the given facts**

Reserve the independent task

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

Investigate, record the evidence and check the limits

> Distinguish a prediction, a paper trace and an actual local execution.

Current investigation

Method passes 9/10 on current sources and 5/10 on outdated sources. Show the intermediate steps and give the conclusion with its scope.

Complete investigation instructions

1. Predict the main 90% and 50% rates and forty-point gap. The complete code separately computes the picture's 87.5%, 37.5% and pooled 62.5%, making the different denominators visible. It summarizes supplied counts only; it does not run a classifier or verify the declared controls. Save the slice results alongside the pooled value and propose a follow-up that holds the baseline, metric and eligibility aligned.

   Leave causal explanations and population rates unresolved. Keep the independent baseline/candidate comparison for the first individual attempt.

```
"""Condition-specific descriptive rates from stipulated summaries."""
from fractions import Fraction

def rate(correct,total):
 if type(correct) is not int or type(total) is not int or total<1 or not
0<=correct<=total:raise ValueError('valid count and positive denominator required')
 return Fraction(correct,total)
main_current=rate(9,10);main_outdated=rate(5,10)
indoor=rate(7,8);outdoor=rate(3,8);pooled=rate(10,16)
assert 100*(main_current-main_outdated)==40
assert 100*(indoor-outdoor)==50 and pooled==Fraction(5,8)
# Reserved independent case is deliberately not solved by this practice program.
print('Main current/outdated rates:',main_current,main_outdated,'gap in percentage
points:',100*(main_current-main_outdated))
print('Separate indoor/outdoor rates:',indoor,outdoor,'pooled:',pooled)
print('Pooled 62.5% hides indoor 87.5% versus outdoor 37.5%; it is not the rate for either
slice.')
print('No individual case outcomes, cause, significance, stable population rate or
universal robustness is supplied. Frozen-model/rubric claims are exercise assumptions, not
code-verified controls.')
```

```
print('No model is run and no new condition is collected. Keep baseline, metric,
eligibility and conditions aligned in any proposed follow-up.')
```

Evidence record and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try the selected new case independently

> Compare baseline and candidate within each condition, preserving opposite changes and the

limits of aggregate counts.

Complete independent task

Baseline/candidate on condition A:6/8, 7/8; condition B:5/8, 4/8. Compare paired rates.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

Attempt this without hints

Baseline/candidate on condition A:6/8, 7/8; condition B:5/8, 4/8. Compare paired rates.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save the research artifact and unresolved conditions

> Keep supplied evidence, proposed revisions and actual work distinct.

Project task and artifact

Add a labelled robustness across conditions **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of robustness across conditions.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence and limitations

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Retrieve the mechanism and state the evidence boundary

> Explain what follows and what would require new evidence.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of robustness across conditions

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use research vocabulary precisely**

REFERENCE · SECTION 5

Reference · Use research vocabulary precisely

> Connect each term to the question, comparison, measurement or evidence.

Vocabulary and your explanation

Word	Meaning
Robustness check	Evaluation under declared varied conditions
Condition-specific result	A score within one stated setting
Aggregate masking	A summary hiding differing condition outcomes

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

- A report omits this qualification: Robustness is bounded by the tested variations and independent **evidence** policy. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the research reasoning and its limits

REFERENCE · SECTION 12

Reference · Check the research reasoning and its limits

> Use after practice; reserve the selected independent case for its first attempt.

Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Robustness across conditions — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Method passes 9/10 on current sources and 5/10 on outdated sources.	Keep baseline, metric and eligibility aligned across conditions. Report condition counts and failures separately; new conditions may extend evidence but cannot guarantee all conditions.	90% versus 50%; gap 40 percentage points. The outdated-source condition is associated with lower quality here; inspect mechanisms before general causal claims.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ Reference · The earlier route checklist

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Check robustness across conditions. A fixed classifier is tested separately on indoor and shaded outdoor cases.

By the end, I can	How I will show it
Explain the mechanism and terminology of robustness across conditions	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Check robustness across conditions. A fixed classifier is tested separately on indoor and shaded outdoor cases.
- Which labelled step matches this detail? The large observed gap motivates investigation rather than claiming condition independence.
- 1. Name both slices
- 2. Compute rates
- 3. Bound robustness
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Bound robustness The large observed gap motivates investigation rather than claiming condition independence.

Robustness asks whether behaviour survives relevant changes in conditions. A total score can hide a weak slice; keep the scenario and denominator visible.

Changed-case question and optional hint

- Changed-case practice: Report only the combined 10/16=62.5%. Which conclusion is supported by the supplied facts?
- 1. The arithmetic is correct, but the condition gap becomes hidden.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The arithmetic is correct, but the condition gap becomes hidden. Combining records discards the slice distinction relevant to the deployment goal.

Repair and retain the record

Review Inspecting failures and anomalies (ai-7/26). Goal: Explain the **mechanism** and terminology of inspecting failures and anomalies. Short practice: Calculate mean and median for 2,2,2,10.

What does the timeout log establish, and why is deleting that inconvenient record incompatible with the stated plan?. Return to this week and explain what you changed.

Review Reproducible environment and records (ai-7/24). Goal: Explain the **mechanism** and terminology of reproducible environment and records. Short practice: Trace the recorded mean for 2,4,6 and its changed result for 2,4,9.

Which artefacts must accompany the **output** so another team can identify the exact run?. Return to this week and explain what you changed.

Review Predefining an analysis plan (ai-7/21). Goal: Explain the **mechanism** and terminology of predefining an analysis plan. Short practice: Apply the recorded exclusion rule.

Can two readable model failures be removed, and how must a necessary later plan change be disclosed?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work
</details>

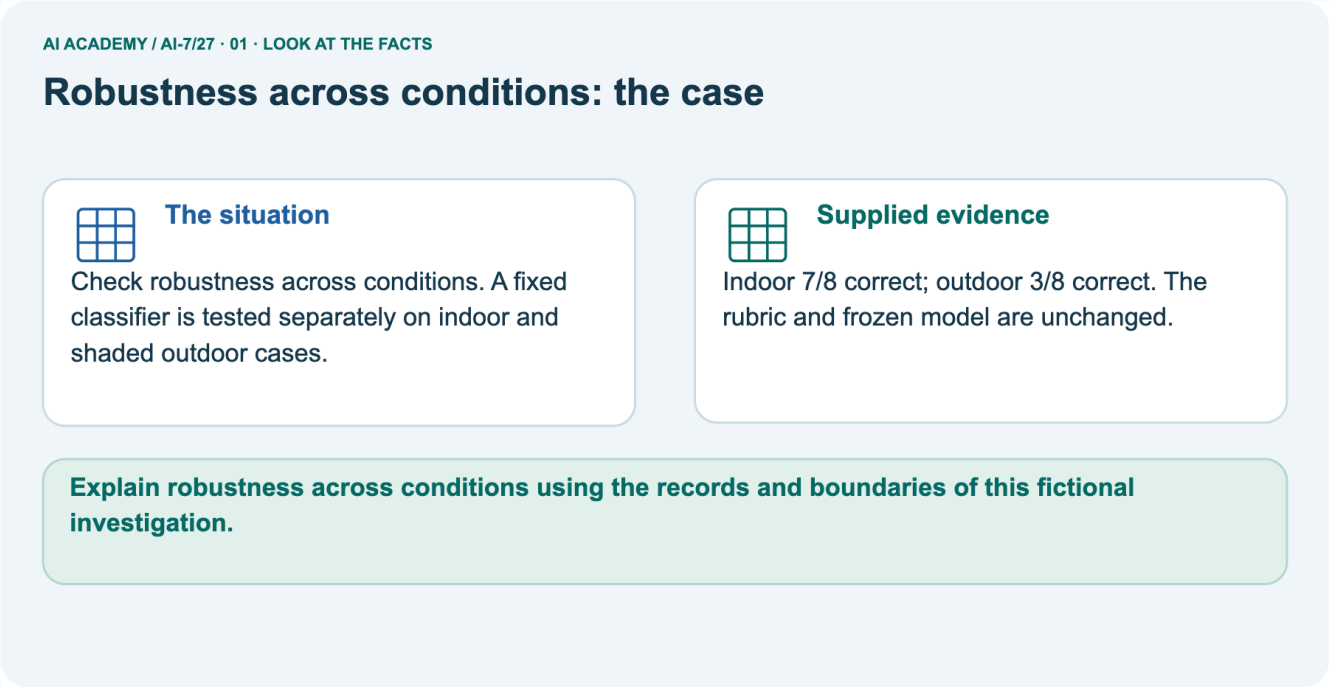
<details>
<summary>Reference · Six-picture concept journey</summary>
</details>

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Robustness across conditions: the case

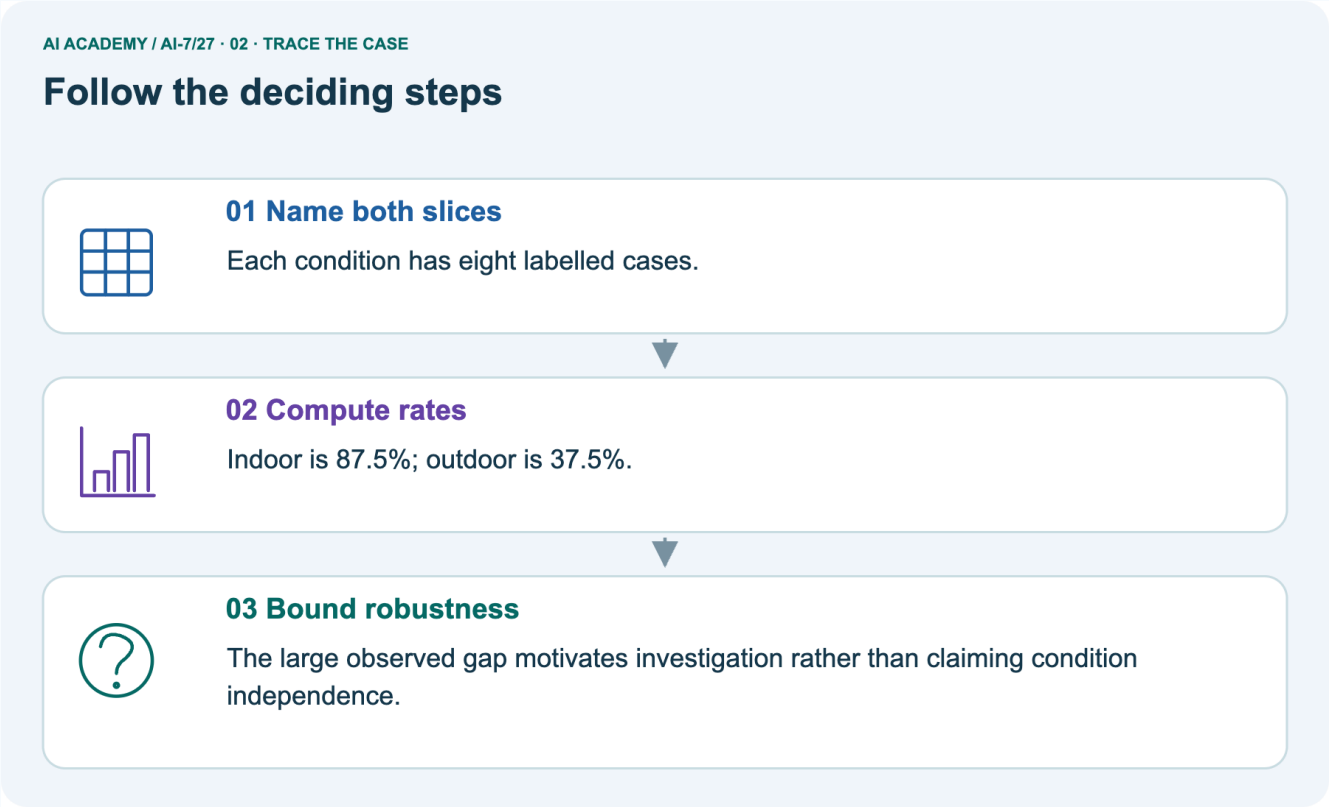


Follow the labelled picture with the matching explanation. Open the vector diagram

Check robustness across conditions. A fixed classifier is tested separately on indoor and shaded outdoor cases. Supplied facts: Indoor 7/8 correct; outdoor 3/8 correct. The rubric and frozen model are unchanged.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

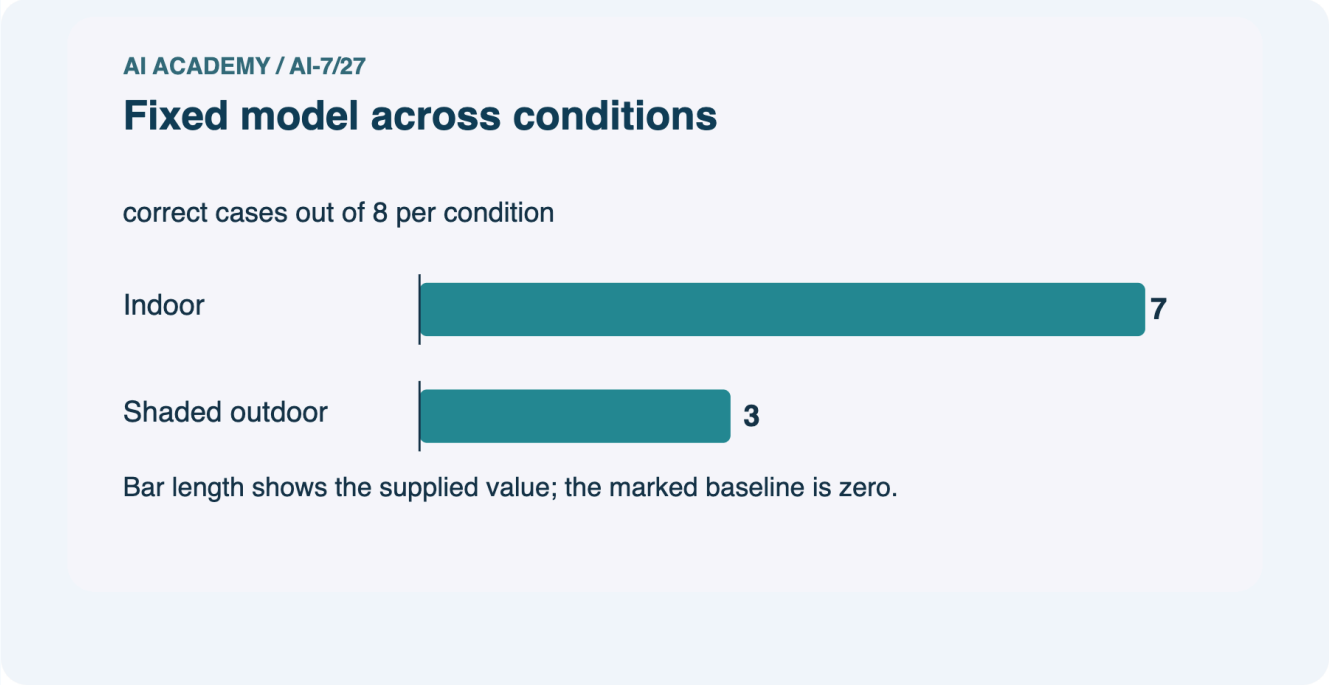


Follow the labelled picture with the matching explanation. Open the vector diagram

Name both slices: Each condition has eight labelled cases. → Compute rates: Indoor is 87.5%; outdoor is 37.5%. → Bound robustness: The large observed gap motivates investigation rather than claiming condition independence.

Point to each numbered stage and explain which recorded **input** it uses.

03. Fixed model across conditions

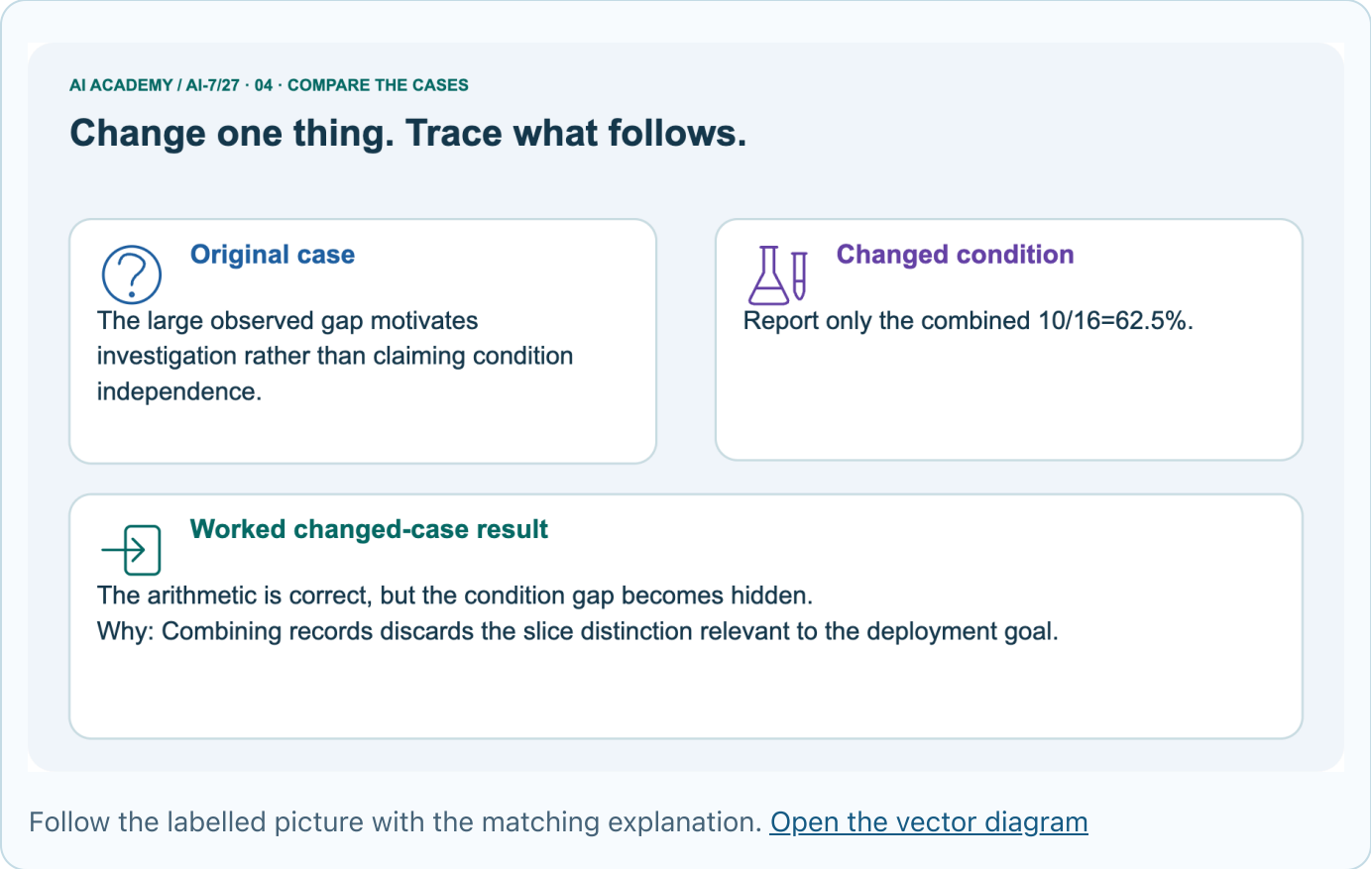


Follow the labelled picture with the matching explanation. Open the vector diagram

Fixed model across conditions. Units: correct cases out of 8 per condition. Indoor: 7. Shaded outdoor: 3.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



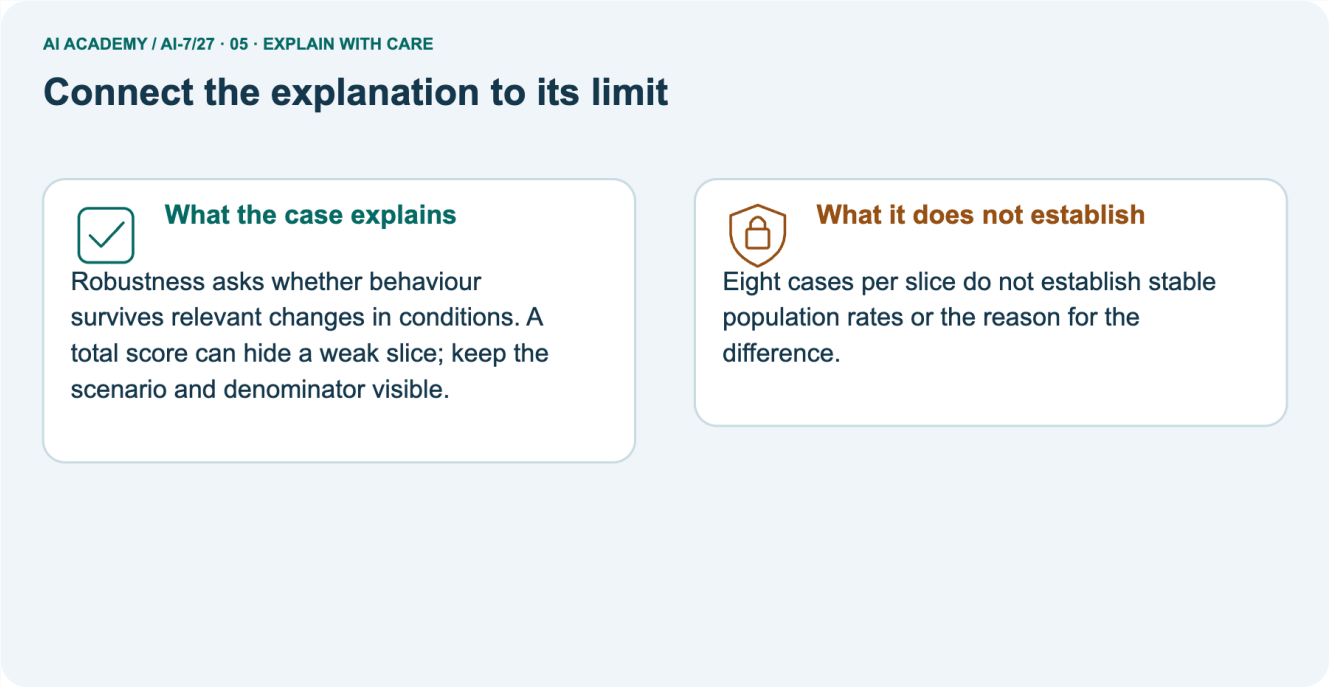
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The large observed gap motivates investigation rather than claiming condition independence. Changed condition: Report only the combined 10/16=62.5%.

Expected result: The arithmetic is correct, but the condition gap becomes hidden. Reason: Combining records discards the slice distinction relevant to the deployment goal.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



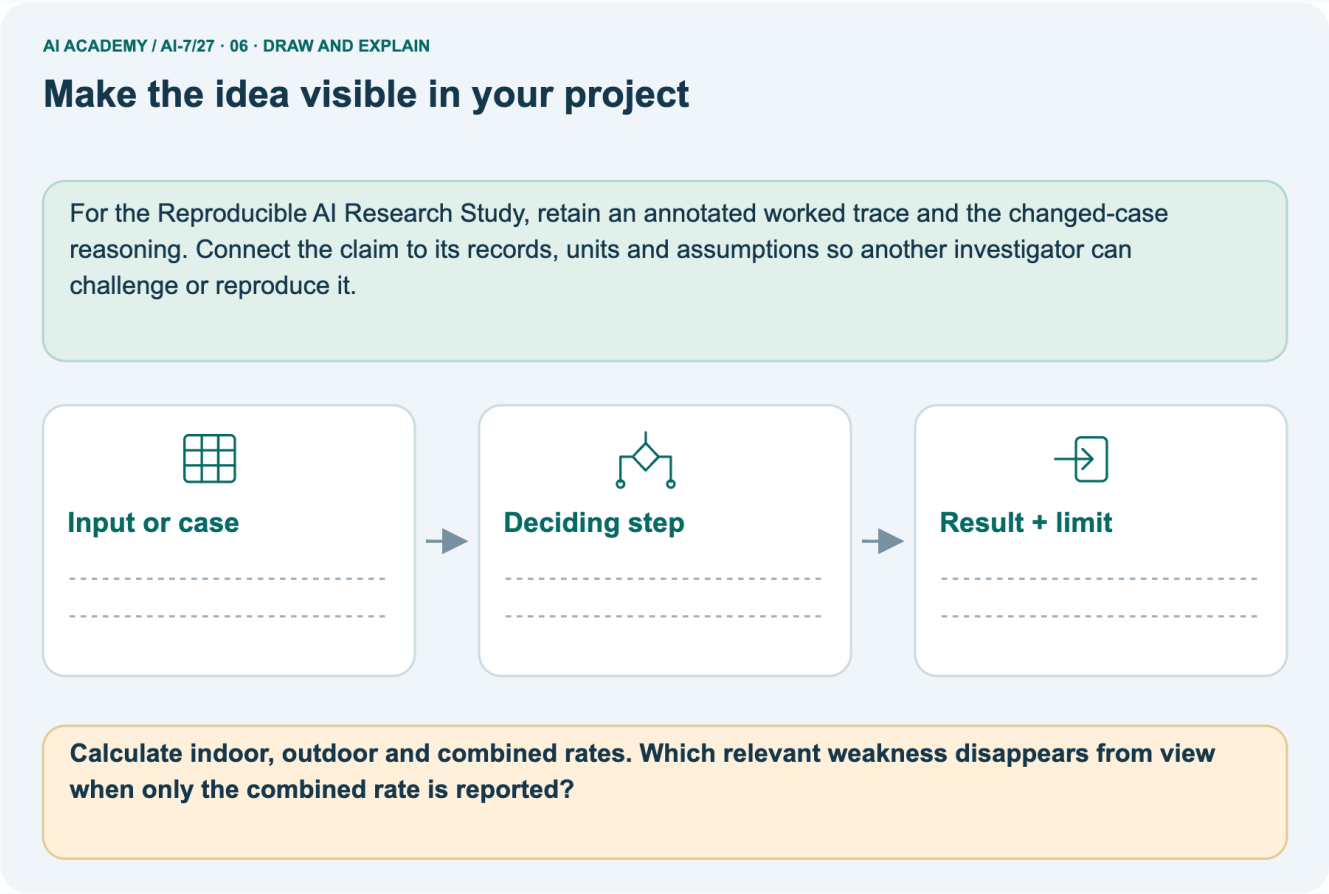
Follow the labelled picture with the matching explanation. Open the vector diagram

Robustness asks whether behaviour survives relevant changes in conditions. A total score can hide a weak slice; keep the scenario and denominator visible.

Limit: Eight cases per slice do not establish stable population rates or the reason for the difference.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate indoor, outdoor and combined rates. Which relevant weakness disappears from view when only the combined rate is reported?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Robustness across conditions: the case

02. Follow the deciding steps

03. Fixed model across conditions

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.